

Mathematical Analysis 2

Metric spaces 1

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1 Examples of metric spaces

The following definitions give metric spaces (M, d) . Verify this!

(a) **Absolute value.**

$$M = \mathbb{R}, \quad d(x, y) = |x - y|.$$

For parts (b)–(e), let $M = \mathbb{R}^n$, where $n \in \mathbb{N}$ (with $n = 2$ in part (e)).

(b) **Euclidean metric.**

$$d_2(x, y) = \sqrt{\sum_{i=1}^n |x_i - y_i|^2}.$$

The Cauchy–Bunyakovsky–Schwarz inequality is

$$\sum_{i=1}^n a_i b_i \leq \sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}.$$

Generalization:

$$d_p(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}, \quad p \in \mathbb{R}, \quad p \geq 1.$$

(c) **Manhattan (sum) metric.**

$$d_1(x, y) = \sum_{i=1}^n |x_i - y_i|.$$

(d) **Maximum metric.**

$$d_{\max}(x, y) = \max\{|x_i - y_i| : 1 \leq i \leq n\}.$$

(e) **Paris metric** ($n = 2$). If $x, y \in \mathbb{R}^2$ lie on the same ray from the origin $(0, 0)$, set

$$d_{\text{par}}(x, y) = d_2(x, y).$$

Otherwise, set

$$d_{\text{par}}(x, y) = \|x\| + \|y\|,$$

where $\|\cdot\|$ is the Euclidean norm.

(f) **Discrete metric.** Let $M \neq \emptyset$ and define

$$d(x, y) = \begin{cases} 1, & x \neq y, \\ 0, & x = y. \end{cases}$$

- (g) **The space of bounded functions with the supremum metric.** Let $M = F(a, b)$ be the set of all bounded real-valued functions on the interval $[a, b]$, where $a, b \in \mathbb{R}$ and $a \leq b$. Define

$$d_{\sup}(f, g) = \sup\{|f(x) - g(x)| : x \in [a, b]\}.$$

- (h) **Distance in an undirected connected graph.** Let $G = (V, E)$ and $M = V$. Define $d(u, v)$ to be the number of edges in a shortest path between u and v .
- (i) **Edit distance.** Let Σ be a finite alphabet of symbols, for example $\Sigma = \{a, b, c\}$. Words are finite strings of symbols; the empty word, denoted by λ , is also a word. We say that two words differ by one edit if one can be obtained from the other by:
- deleting one symbol anywhere in the word;
 - inserting one symbol anywhere in the word;
 - replacing one symbol by another at a given position in the word.

The distance (metric) between two words is the smallest number of edits needed to transform one word into the other.

What is a neighbourhood? What are open and closed sets? What is a ball of radius r ?

2 Unit balls

What does the unit ball, that is, the set

$$B(a, 1) = \{x \in \mathbb{R}^n : d(a, x) < 1\},$$

look like in $\mathbb{R}^n$ with each of the following metrics?

- the Euclidean metric;
- the Manhattan (sum) metric;
- the maximum metric;
- the Paris metric (in $\mathbb{R}^2$).

3 Strongly equivalent metrics

Definition: strong equivalence

Two metrics d_i, d_j on the same set M are called **strongly equivalent** if there exist $r, s \in (0, \infty)$ such that

$$r d_i(x, y) \leq d_j(x, y) \leq s d_i(x, y) \quad \text{for all } x, y \in M.$$

Show that the Euclidean, Manhattan, and maximum metrics are strongly equivalent.